GARRICK JOHN JENSEN

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EDUCATION

University of California, Los Angeles Expected June 2028
Master of Science, Mechanical Engineering

Harvey Mudd College May 2022
Bachelor of Science, General Engineering | Graduated with Distinction, Dean's List

TECHNICAL SKILLS

Languages & Tools: Python, MATLAB, C/C++ (coursework), Linux, Git; PyTorch, Scikit-learn, SciPy, OpenCV
Robotics & Perception: Computer vision, segmentation, object detection, pose estimation, sensor fusion, state estimation (EKF, particle filters), occupancy mapping, path planning (RRT, coverage)
Learning & Signals: Deep learning, transformers, CNNs, reinforcement learning, world models, multi-agent RL, GPU-accelerated simulation, Fourier analysis, FMCW radar, LiDAR, EO/IR

RESEARCH AND PROFESSIONAL EXPERIENCE

UCLA Robotics VECTR Lab, Los Angeles, CA September 2026–Present
Graduate Student Researcher

- Developing a GPU-accelerated multi-agent RL environment for cooperative search in GPS- and comms-denied conditions, with simulated LiDAR and occupancy mapping under pose uncertainty
- Implementing and comparing model-based (DreamerV3) and model-free (IPPO) multi-agent learning arms
- Teaching assistant for the Mechanical and Aerospace Engineering undergraduate capstone course

Areté Associates, Northridge, CA August 2022–December 2026
Scientist 2

- Developed computer vision algorithms and deep learning models for an autonomous triaging device for mass casualty events
- Conducted field tests in military exercises and national competitions such as the DARPA Triage Challenge; presented work at the Military Health System Research Symposium
- Beginning to end training and validation of a joint human pose, body part, and skin segmentation model for real time deployment in austere military conditions
- Signal processing in both the real and Fourier space for heart and respiration rate extraction from FMCW radar and IR/EO imagery for a multimodal vitals monitor
- Design of a custom 2D transformer based classification model for efficient detection of 3D targets in undersea LiDAR, outperforming traditional 3D ResNeXt models with a 0.97 AUC
- Trained RL agents deployed on drones for navigating high dimensional spaces under time constraint

Doosan Bobcat, West Fargo, ND August 2021–May 2022
Capstone Team Lead and Robotics Engineer

- Developed a coverage path planning algorithm using boustrophedon curves optimized as a modified traveling salesman problem
- Implemented an Extended Kalman Filter fusing radar and GPS for localization, plus a particle filter for obstacle detection and avoidance, on an autonomous mower
- Led a five-person team: set schedule, budget, and tasking, and served as primary liaison to the company sponsor

SELECTED PROJECTS

RRT Path Planning for Obstacle Avoidance and Rendezvous. Rapidly-exploring Random Trees planning to a moving target through cluttered space.

Two-Segment Robotic Arm. Dynamically modeled and built a stepper-driven planar arm; programmed an STM32 Nucleo-64 and Max1000 FPGA over SPI and UART to trace keypad-entered shapes.

PATENT AND SELECTED POSTER PRESENTATIONS

Patent Pending (named inventor). Schmidt, M., et al. "Path Planning for Automatic Mowers." WO2023205244A1, Oct. 2023.

Jensen, G., Sato, R., Smith, R., Gushurst, W. "Reinforcement Learning Based Path Planning for Injury-Aware Mass Casualty Monitoring." Military Health System Research Symposium (MHSRS), Aug. 2026.

Jensen, G., Peel, J., Roman, K. "Adapting Open-Source Artificial Intelligence to Combat Casualty Care: Joint Human and Skin Segmentation Neural Network." MHSRS, Aug. 2024.

Four additional co-authored poster presentations at MHSRS (2023, 2024) and BMES (2019).